

Response to Reviewers

Submission 19

Sustainability Context Inheritance: From Trusted Evidence to Product Sustainability Profiles in Circular Economy Decision-Making

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Thank you to the reviewers for their constructive feedback. The camera-ready revision preserves the accepted conceptual / work-in-progress scope while clarifying the decision context, implementation logic, evidence qualification, sufficiency assessment, comparative positioning and illustrative evaluation. The responses below address each comment individually and identify the corresponding revisions in the manuscript.

Reviewer 1

Comment 1.1

Reviewer comment: *Clarify intended decision contexts (e.g., consumers, OEMs, regulators) and target user groups.*

Response: The intended users of SCIA and the meaning of a Decision Context are made explicit in the revised manuscript. Example decision roles include OEMs, regulators, financial institutions, downstream actors and recyclers. A Decision Context specifies who is making or consuming the assessment and the purpose for which the sustainability claim is being evaluated. The Applicable Policy then determines the evidence requirements relevant to that decision, including admissibility conditions, materiality weights, mandatory conditions and thresholds.

Revision in manuscript: Section 1 defines Decision Context and identifies the intended decision roles. Figure 1 shows Decision Context and Applicable Policy as inputs to the Context-Specific Inheritance Assessment. Section 7 provides a concrete OEM decision example and notes that another actor, such as a recycler, may reach a different outcome from the same underlying evidence when different evidence paths and policy conditions apply.

Comment 1.2

Reviewer comment: *SCIA is described at a high level of abstraction, limiting assessment of soundness and feasibility.*

Response: This concern is addressed by making the SCIA processing flow, evidence representation and plausible implementation path more explicit while retaining the conceptual scope of the paper. The revised architecture identifies the evidence inputs, generic evidence-state constructs, Decision Context and Applicable Policy, Context-Specific Inheritance Assessment, and the resulting Product Sustainability Profile. A more concrete representation of the Sustainability Lineage Graph and the mechanisms through which distributed evidence could be resolved, matched and retained for assessment are also provided.

Revision in manuscript: Section 4 and Figure 1 make the SCIA execution flow and the relationship among its principal constructs explicit. Section 5 defines the Sustainability Lineage Graph as $G=(V,E)$, identifies representative node and relation types, and specifies evidence qualification information such as source/issuer, scope, provenance, validity and assurance basis. Section 5 further provides a technology-agnostic implementation path in which evidence is resolved from available sources and matched to product, component, process or lifecycle nodes using identifiers and semantic mappings. GS1 Digital Link / Resolver and EPCIS are identified as possible discovery and lineage-information mechanisms, while RDF or property graphs are presented as possible representations rather than mandatory technologies. Section 3 clarifies that the current evaluation remains conceptual and that prototype performance and empirical validation are future work.

Comment 1.3

Reviewer comment: *Expand the electric scooter use case; provide concrete attribution rules and additional examples.*

Response: The electric-scooter use case has been expanded from a descriptive illustration into a worked decision example with explicit evidence paths, quantitative coverage and gap calculations, policy thresholds, and a mandatory admissibility condition. The revised example distinguishes between aggregate evidence coverage and decision-specific admissibility, showing that a product-level outcome cannot be derived from coverage alone. Within the page-limited scope of the paper, additional variation is illustrated through multiple component and lifecycle evidence paths and through a contrasting decision context in which another actor may apply different evidence requirements to the same underlying evidence.

Revision in manuscript: Section 6 evaluates battery manufacturing, aluminium/frame production, electronics manufacturing, final assembly and tyres/other upstream as separate evidence paths with policy-defined materiality weights and coverage values. Table 2 produces C_D=82.5% and G_D=17.5%, and applies illustrative thresholds for Supported, Partially Supported and Unsupported outcomes. Section 7 then applies a mandatory battery-assurance condition: although the battery evidence is present, traceable and fully covered, the required verification has expired, resulting in an Unsupported final PSP outcome. Table 3 records the claim, decision, applicable policy, evidence state, blocking condition, coverage gaps and required action. A recycler assessing end-of-life recoverability is also identified as an example in which the same underlying evidence may be evaluated using different evidence paths and policy conditions.

Comment 1.4

Reviewer comment: *Provide more insight into technologies/mechanisms for checking and matching processes central to SCIA.*

Response: A clearer implementation-oriented sequence has been added for the checking and matching processes underlying SCIA. The Sustainability Lineage Graph provides the target semantic structure through defined node types (such as Product, Component, Material, LifecycleStage, Evidence and Credential) and relations (such as contains, sourcedFrom, processedAt, evidencedBy and appliesTo). Incoming evidence can then be resolved from available sources and matched to the relevant graph nodes and evidence paths using identifiers and semantic mappings. Provenance and qualification metadata are retained with the matched evidence so that its source, scope, validity and assurance status remain traceable during subsequent assessment. Existing standards and mechanisms are treated as complementary infrastructure rather than prescribed SCIA technologies.

Revision in manuscript: Section 5 makes the matching mechanism explicit through the Sustainability Lineage Graph structure. The defined node and relation types provide the semantic structure against which incoming evidence is associated. Evidence retrieved from distributed sources is matched to the relevant product, component, process or lifecycle nodes using identifiers and semantic mappings, while provenance and qualification metadata are retained to preserve traceability and support later admissibility assessment. GS1 Digital Link / Resolver is identified as a possible mechanism for resolving distributed evidence resources, and EPCIS as a possible source of visibility-event information for establishing lineage. RDF or property graphs are presented as possible representations of this semantic and lineage structure. Section 2 and Table 1 position DPPs, EPCIS, Digital Link / Resolver and trust/conformity mechanisms as complementary evidence, discovery, lineage and assurance sources.

Comment 1.5

Reviewer comment: *Discuss validation of information artifacts used as input.*

Response: The treatment of input-artifact validation has been further clarified by making the evidence-qualification information more explicit. For each artifact, the assessment retains information needed to understand its source, scope and applicability to the relevant product or constituent path. This includes provider or issuer information, product/component scope, provenance, validity or expiry status, and the verification or assurance basis where available. Source-specific validation or verification results are preserved as part of the evidence qualification rather than being independently recreated by SCIA.

Revision in manuscript: Section 5 provides additional detail on the qualification metadata associated with evidence represented in the Sustainability Lineage Graph. Evidence or Credential nodes can be linked to the relevant product/component path while retaining provider/issuer, scope, provenance, validity/expiry and verification or assurance information. This enables the assessment to distinguish evidence that is available from evidence that is applicable and usable for the particular decision. The section also clarifies that SCIA relies on source-specific

validation or verification mechanisms where these exist and carries their results forward as qualification metadata for subsequent admissibility assessment.

Comment 1.6

Reviewer comment: *Discuss verification of authenticity and trustworthiness of sustainability certificates.*

Response: The distinction between certificate authenticity, assurance information and decision-specific admissibility has been clarified. SCIA does not act as an independent certificate-verification or trust mechanism. Instead, it retains available qualification and assurance information associated with the evidence, such as issuer or provider, signer where applicable, validity or expiry status, and verification or assurance basis. Where authenticity or verification is established by an external credential, certification, audit or assurance mechanism, that result becomes part of the evidence qualification available to SCIA. Whether such evidence is acceptable for a particular claim is then determined by the Applicable Policy.

Revision in manuscript: Sections 2 and 5 clarify the complementary role of credential and trust/conformity mechanisms and the qualification information retained with evidence. Section 6 further distinguishes evidence coverage from admissibility by noting that a policy may require independent assurance rather than self-attestation, a recognized issuer or scheme, an accepted assessment methodology, or a credential that is valid on the assessment date. Section 7 illustrates this distinction through the battery-manufacturing evidence: the evidence remains present and traceable, but the required independent verification has expired, so the mandatory admissibility condition is not satisfied and the final PSP outcome is Unsupported.

Comment 1.7

Reviewer comment: *Define and justify the methods used to assess evidence sufficiency.*

Response: The evidence-sufficiency method is articulated more explicitly as a decision-scoped assessment rather than a universal sustainability score. Relevant evidence paths are evaluated using evidence coverage c_i and policy-defined materiality weights w_i . These are used to derive decision-relevant coverage C_D , individual evidence gaps g_i , and the total Sustainability Gap G_D . The method further separates evidence availability from policy-specific admissibility: coverage indicates how much decision-relevant evidence is available, while the Applicable Policy determines whether that evidence may be relied upon and whether any mandatory evidence conditions must be satisfied. What constitutes sufficient evidence may therefore vary by domain, decision role and applicable policy; evidence sufficient for a recycler's end-of-life assessment may not be sufficient for a regulator's compliance decision, even when both assessments use the same underlying evidence base. The resulting inheritance outcome is therefore determined from both quantified evidence sufficiency and the conditions applicable to the particular decision.

Revision in manuscript: Section 6 provides the formal illustrative assessment: $C_D = \sum_i w_i c_i$, $g_i = w_i(1 - c_i)$, and $G_D = \sum_i g_i$, with normalized weights such that $C_D + G_D = 1$. The materiality weights represent the relevance of each evidence path to the specific decision rather than evidence reliability or sustainability performance. The section provides illustrative thresholds for Supported, Partially Supported and Unsupported outcomes and clarifies that these thresholds and weights are policy-defined rather than universal SCIA constants. It also distinguishes three assessment questions: whether relevant evidence is available, whether it is admissible under the Applicable Policy, and whether the admissible evidence is sufficient for product-level inheritance. Section 7 demonstrates the decision-specific nature of sufficiency through the OEM assessment and a contrasting recycler example, noting that the same evidence may lead to a different outcome when different evidence paths, weights, mandatory conditions and thresholds apply.

Reviewer 2

Comment 2.1

Reviewer comment: *Clarify thresholds, weighting, and uncertainty propagation.*

Response: The treatment of thresholds, weighting and uncertainty has been further clarified. Materiality weights are defined by the Applicable Policy according to the relevance of each evidence path to the particular claim and Decision Context; they do not represent evidence reliability or sustainability performance. Similarly, the thresholds

used to derive Supported, Partially Supported and Unsupported outcomes are policy-specific rather than universal SCIA constants. The values used in the electric-scooter scenario are therefore illustrative parameters for demonstrating the assessment method rather than empirically calibrated thresholds. Consequently, both materiality weighting and sufficiency thresholds may differ across decision contexts; the evidence requirements appropriate for an OEM, regulator or recycler need not be identical. Unresolved evidence is retained explicitly rather than being absorbed into the available evidence. Individual gap terms g_i and the aggregate Sustainability Gap G_D quantify decision-relevant evidence deficits, while mandatory evidence requirements allow critical deficiencies to remain visible even when aggregate coverage is relatively high. Formal probabilistic uncertainty propagation is outside the scope of the current conceptual paper and is identified as a direction for future work.

Revision in manuscript: Section 6 defines c_i as evidence coverage for a relevant path and w_i as its policy-defined materiality weight, with $C_D = \sum_i w_i c_i$, $g_i = w_i(1-c_i)$, and $G_D = \sum_i g_i$. The text clarifies that the weights reflect decision relevance rather than evidence reliability and that the illustrative support thresholds are supplied by the Applicable Policy rather than prescribed universally by SCIA. Sustainability Gaps remain explicit through g_i and G_D , and Section 7 further shows that a mandatory evidence condition can override an aggregate coverage result so that a critical deficiency is not averaged away. Section 8 identifies calibration and formal uncertainty propagation as future work.

Comment 2.2

Reviewer comment: *Improve the resolution of Figures 1-3.*

Response: The presentation of Figures 1-3 has been refined for publication-size readability. Higher-resolution figure assets are used, with simplified visual layouts, clearer labels and reduced information density so that the figures remain legible when reproduced at LNI page size.

Revision in manuscript: Figures 1-3 use the revised publication-ready assets. Figure 1 presents the SCIA architecture and assessment flow; Figure 2 presents the Sustainability Lineage Graph and the distinction between evidenced, partially evidenced and missing/unknown paths; and Figure 3 provides a simplified view of federated sustainability evidence contributing to a decision-scoped PSP. The figures have been sized consistently within the manuscript and simplified to preserve readability in print and PDF viewing.

Comment 2.3

Reviewer comment: *Benchmark / position SCIA against EPCIS, GS1 Digital Link and Digital Product Passport mechanisms.*

Response: The relationship between SCIA and existing product-information and traceability mechanisms is positioned more explicitly through a role-based comparison. Digital Product Passports can provide structured product and lifecycle information and associated sustainability evidence; GS1 EPCIS can provide cross-organizational event and traceability information supporting lineage; and GS1 Digital Link / Resolver can support identifier-based discovery of distributed information resources. These mechanisms address complementary parts of the information infrastructure that SCIA can consume. SCIA is positioned at the subsequent decision-oriented layer, where relevant evidence and lineage are interpreted in relation to a particular Decision Context and Applicable Policy to assess evidence coverage, gaps, admissibility and product-level attribution. The paper therefore does not position SCIA as a replacement for DPP, EPCIS or Digital Link / Resolver. Within the conceptual scope of the present paper, this comparison is intended as functional positioning; experimental interoperability or performance benchmarking across these mechanisms remains outside the current evaluation scope.

Revision in manuscript: Section 2 discusses the complementary roles of DPPs, GS1 EPCIS, GS1 Digital Link / Resolver and digital credential or trust/conformity mechanisms before defining the specific problem addressed by SCIA. Table 1 provides a direct comparison by primary role and relationship to SCIA: DPPs are identified as potential sources of structured product/lifecycle sustainability evidence; EPCIS as a potential event and lineage source; Digital Link / Resolver as an optional discovery mechanism; and trust/conformity mechanisms as sources of qualification and assurance information. SCIA is distinguished by its focus on decision-scoped evidence sufficiency and attribution, producing a PSP from relevant evidence, coverage, gaps and applicable rules.

Comment 2.4

Reviewer comment: *Strengthen quantitative evaluation / decision-quality assessment.*

Response: The illustrative evaluation has been made more quantitative while retaining the conceptual/WIP scope of the paper. The electric-scooter assessment now provides explicit materiality weights, evidence-coverage values, quantified Sustainability Gaps and policy thresholds, making the derivation of the assessment outcome reproducible from the stated illustrative inputs. The subsequent admissibility assessment demonstrates a second form of decision discrimination: a coverage-based result can change when a mandatory evidence condition is not satisfied. This allows the framework's internal decision logic and transparency to be examined quantitatively without claiming empirical improvement in human decision quality. Evaluation of actual decision quality requires real-product evidence and stakeholder or controlled evaluation and is therefore treated as future work.

Revision in manuscript: Section 6 and Table 2 provide the quantitative coverage and gap assessment, yielding C_D=82.5% and G_D=17.5% for the illustrative scooter case and a coverage-based Partially Supported outcome. Section 7 then evaluates the applicable mandatory assurance condition and shows that the battery path can be fully represented in coverage terms while still being inadmissible because the required verification has expired; the final PSP outcome is therefore Unsupported. Table 3 makes the decision result transparent and traceable by reporting the claim, decision, Applicable Policy, observed evidence state, blocking condition, coverage gaps and required action. Section 3 identifies the present evaluation as conceptual, and Section 8 identifies externally sourced product evidence, real-product prototyping, calibration and stakeholder evaluation as the next empirical phase.

Overall Revision

The camera-ready revision addresses the substantive reviewer requests while preserving the accepted conceptual contribution and short-paper scope. The principal refinements are clearer Decision Context and Applicable Policy, more explicit implementation and evidence-qualification logic, formal decision-relevant coverage and gap calculations, explicit admissibility and mandatory conditions, a more quantitative electric-scooter assessment, refined positioning against complementary standards and infrastructures, improved figure readability, and a clearer empirical-validation path for future work.